Cognitive Modeling

Complete Lecture Notes

Week 7: Evaluating Models II: Information Criteria, Parameter Recovery and Model Recovery

Contents

1 Evaluation of Cognitive Models 1

1.1 Week 7: Evaluating Models II: Information Criteria, Parameter Recovery and Model Recovery . . . 1

1.1.1 Key terms . . . 1

1.1.2 Learning outcomes . . . 2

1.1.3 Motivation & big picture . . . 3

1.1.4 Lecture notes . . . 3

1.1.5 Worked example . . . 8

1.1.6 Lab (2 hours) . . . 9

1.1.7 Connection to AI & LLMs . . . 12

1.1.8 Common mistakes & misconceptions . . . 13

1.1.9 Check your understanding . . . 13

1.1.10 Assignment . . . 14

1.1.11 Further reading . . . 14

Chapter 1

Evaluation of Cognitive Models

1.1 Week 7: Evaluating Models II: Information Criteria, Parameter Recovery and Model Recovery

> Chapter 3: Evaluation of cognitive models · Lecture 1 h · Lab 2 h Running data: the Week 2–6 forgetting data and models, plus large simulation studies at the same design.

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
criterion (c)	ملاک	آستانه‌ی تصمیم؛ نشان‌دهنده‌ی سوگیری پاسخ
parameter recovery	بازیابی پارامتر	آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
model recovery	بازیابی مدل	آیا مقایسه‌ی مدل، مدل مولد را درست انتخاب می‌کند
forgetting curve	منحنی فراموشی	کاهش یادآوری با گذشت زمان
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
research design	طرح آزمایش	شرایط، تعداد کوشش‌ها و نمونه‌ها

English	فارسی	توضیح کوتاه
likelihood-ratio test (LRT)	آزمون نسبت درست‌نمایی	مقایسه‌ی دو مدل تودرتو
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
nested models	مدل‌های تودرتو (nested)	یک مدل حالت خاص مدل دیگر است
AIC / BIC	بیزی / معیار اطلاعاتی آکائیکه	برازش به‌علاوه‌ی جریمه برای تعداد پارامتر
response bias	سوگیری پاسخ	تمایل به یک پاسخ، مستقل از محرک
confusion matrix	ماتریس درهم‌ریختگی (confusion matrix)	P(مدل انتخاب‌شده)
inversion matrix	ماتریس معکوس (inversion matrix)	P(مدل مولد)
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
exponential / power function	توانی / تابع نمایی	دو شکل رایج برای منحنی فراموشی و یادگیری
identifiability	شناسایی‌پذیری	آیا داده پارامترها را به‌طور یکتا تعیین می‌کند
parameter trade-off	بده‌بستان پارامترها	تغییر یک پارامتر که با پارامتر دیگر جبران می‌شود
log-likelihood / NLL	منفی آن / لگاریتم درست‌نمایی	مقداری که بهینه‌سازها کمینه یا بیشینه می‌کنند
Akaike weights	وزن‌های آکائیکه	پشتیبانی نسبی از هر مدل
cross-validation (CV)	اعتبارسنجی متقابل (cross-validation)	برازش روی بخشی از داده و آزمون روی بخش کنارگذاشته
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
model mimicry	تقلید مدل	تولید داده‌ای که شبیه داده‌ی مدل دیگر است
in-context learning	یادگیری درون‌متنی	یادگیری از مثال‌های داخل prompt

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Conduct a likelihood-ratio test for nested models and explain why it cannot compare non-nested models.
2. Compute and interpret AIC, AICc, BIC and Akaike/BIC weights, and explain the different goals of AIC and BIC.
3. Design and run a parameter-recovery study, and diagnose biased, noisy or traded-off parameters.
4. Design and run a model-recovery study, and interpret the confusion and inversion matrices.

5. Evaluate whether a given experimental design and comparison method can support the conclusions a researcher wants to draw.

1.1.3 Motivation & big picture

Cross-validation (Week 6) is general but expensive: every fold means refitting, and refitting complex models with multi-start can take hours. Information criteria estimate out-of-sample performance from a *single* fit, by adding a complexity penalty to the training fit.

But there is a more fundamental question that no criterion answers on its own: can our method even work at this design? Suppose we conclude "participants forget by a power law, with decay exponent 0.42". Two things must be true for this to mean anything:

1. If the data had come from a power model with $d = 0.42$, our fitting procedure would have returned something close to 0.42 (parameter recovery).
2. If the data had come from an exponential model, our comparison method would have said "exponential", not "power" (model recovery).

Both can be checked by simulation, because we know the truth for simulated data. Modern guidelines for computational modeling treat these checks as essential; skipping them is like running an experiment without checking that the equipment works. This week completes the "simulate $\rightarrow$ fit $\rightarrow$ evaluate $\rightarrow$ recover" part of our pipeline.

Where is this used?

- Model comparison tables with AIC/BIC appear in almost every modeling paper in psychology, neuroscience, ecology and economics.
- Computational psychiatry: before claiming that patients have a lower learning rate, researchers must show by parameter recovery that the learning rate can be estimated at their design. Reviewers increasingly require this.
- Designing experiments: recovery simulations are run *before* collecting data, to choose the number of trials and conditions (a modeling version of power analysis).
- Any study that says "model X explains the data best" should report model recovery; otherwise the winner may only reflect a bias of the criterion.

1.1.4 Lecture notes

In plain words Cross-validation is honest but slow. Information criteria (AIC, BIC) are quick formulas that take the fit of each model and subtract a penalty for every parameter, so that a model only

wins if its extra parameters improve the fit by more than they "cost". AIC aims at the best predictor, BIC at the true model, and BIC's penalty is larger. The second, more important half of the week asks whether our methods work at all. We test them on fake data where we *know* the truth. Parameter recovery: simulate data with known parameter values, fit, and check that the fitted values come back close to the true ones. Model recovery: simulate data from each candidate model and check that the comparison method picks the model that actually generated them. If a method cannot find the truth in simulated data, its answer on real data means nothing.

Suggested timing (60 min): likelihood-ratio test 8 · AIC and BIC 20 · parameter recovery 14 · model recovery 14 · identifiability 4.

1. The likelihood-ratio test (nested models)

Model $\mathcal{M}_0$ is nested in $\mathcal{M}_1$ if $\mathcal{M}_0$ is a special case of $\mathcal{M}_1$ obtained by fixing some parameters. Example: the exponential model is the exponential-with-asymptote model with $s = 0$ (asymptote at 0). Under the null hypothesis that $\mathcal{M}_0$ is true,

$$\Lambda = 2\big[\ell_1(\hat{\boldsymbol{\theta}}_1) - \ell_0(\hat{\boldsymbol{\theta}}_0)\big] \mathrel{\dot\sim} \chi^2_{\Delta k},$$

where Δk is the difference in the number of free parameters. A large Λ means the extra parameters improve the fit more than chance would.

Limitations: it only works for nested models (power vs. exponential cannot be tested this way); it is a null-hypothesis test (it can reject the simpler model but never support it); and it is approximate when the fixed parameter value lies on the boundary of the parameter space (as $s = 0$ does).

> Advanced: boundary nulls. When the null value is on the boundary (a variance of 0, an asymptote of 0, a mixture weight of 0), the null distribution of Λ is a 50:50 mixture of χ^2_0 (a point mass at 0) and χ^2_1. Using χ^2_1 is then conservative: the correct p-value is half of the naive one.

2. Information criteria

Information criteria trade off fit against complexity. With maximized log-likelihood $\hat{\ell}$, k free parameters and N observations:

$$\text{AIC} = -2\hat{\ell} + 2k, \qquad \text{AICc} = \text{AIC} + \frac{2k(k+1)}{N-k-1}, \qquad \text{BIC} = -2\hat{\ell} + k \ln N.$$

Lower is better. Only differences between models fitted to the same data are meaningful.

	AIC (and AICc)	BIC
Goal	Best prediction of new data (minimize expected Kullback–Leibler divergence to the truth)	Pick the true model if it is among the candidates (consistency); approximates the Bayes factor
Penalty per parameter	2	$\ln N$ (bigger than 2 once $N \geq 8$)
As $N \to \infty$	May keep choosing over-complex models	Chooses the true model with probability $\to 1$
Small-sample correction	AICc (use when $N/k < 40$)	—

What is N? The number of *independent observations* that enter the likelihood. For binomial recall data that is the number of item responses (here $8 \times 50 = 400$), not the number of cells (8). This choice affects BIC (through $\ln N$) and AICc. Report it.

Weights. With $\Delta_i = \text{IC}_i - \min_j \text{IC}_j$,

$$w_i = \frac{\exp(-\Delta_i/2)}{\sum_j \exp(-\Delta_j/2)}.$$

Akaike weights estimate the probability that model i is the best predictor among the candidates. BIC weights approximate posterior model probabilities under equal prior odds (Week 10).

Rules of thumb for Δ: 0–2, essentially equivalent support; 4–7, considerably less support; more than 10, essentially none. These are conventions, not laws.

Limitation shared by AIC and BIC: both count parameters, k, so they are blind to differences in functional-form flexibility between models with the same k (Week 6). Cross-validation and Bayes factors (Week 10) are not.

> Advanced: where the AIC penalty comes from. The maximized log-likelihood is an optimistically biased estimate of the expected log-likelihood of new data, because the same data were used to choose $\hat{\boldsymbol{\theta}}$. Akaike showed that, for regular models and large N, the bias is approximately k. Correcting for it gives $-2(\hat{\ell} - k) =$ AIC. That is also why AIC and LOO-CV agree asymptotically.

Reading the math

- $-2\hat{\ell}$ is the misfit part (twice the minimized NLL); $2k$ or $k \ln N$ is the penalty part. Lower AIC/BIC is better.
- Each extra parameter must lower $-2\hat{\ell}$ by more than 2 (AIC) or more than $\ln N$ (BIC) to be worth it. With $N = 400$, $\ln N \approx 6$: a parameter must improve the NLL by about 3 units to pass BIC, but only by 1 unit to pass AIC.
- Weights: $\exp(-\Delta_i/2)$ converts a difference in AIC into a relative likelihood. $\Delta = 2$ gives $e^{-1} = 0.37$, so a model 2 AIC units worse is about a third as well supported. Dividing by the

sum makes the weights add up to 1.

- Worked numbers: power model AIC $= 2(19.830)+2(2) = 43.66$; exponential-with-asymptote $= 2(18.919) + 2(3) = 43.84$. The asymptote model fits better (lower NLL) but pays for one more parameter, so the two end up almost tied.

3. Parameter recovery

Question: at this design, does the fitting procedure return the parameters that generated the data?

Procedure: 1. Choose a set of true parameter values $\boldsymbol{\theta}^*$ covering the plausible range (a grid, or random draws from plausible ranges, e.g., around the estimates from real data). 2. For each $\boldsymbol{\theta}^*$, simulate a dataset with exactly the real design (same conditions, numbers of trials). 3. Fit the model with exactly the real fitting procedure (same optimizer, starts, bounds). 4. Compare $\hat{\boldsymbol{\theta}}$ with $\boldsymbol{\theta}^*$: scatter plots of true vs. recovered; correlation; bias (mean of $\hat{\theta} - \theta^*$); RMSE. 5. Also plot the recovered parameters against each other. A strong correlation between, say, $\hat{\lambda}$ and $\hat{s}$ that is absent in the true values reveals a trade-off.

Reading the results: points on the identity line mean good recovery; a cloud parallel to the x-axis means the parameter is not recoverable at this design; a curve bending away at the extremes means bias near the bounds. If recovery fails, do not interpret that parameter. Change the design (more trials, other conditions) or simplify the model.

4. Model recovery

Question: if model $\mathcal{M}_a$ generated the data, does our comparison method select $\mathcal{M}_a$?

Procedure: 1. For each candidate model $\mathcal{M}_a$: simulate many datasets at the real design, with parameters drawn from plausible ranges. 2. Fit every candidate model to every dataset. 3. Select a winner by the chosen criterion (AIC, BIC, CV, ...). 4. Tabulate the confusion matrix $P(\text{selected} = b \mid \text{generated} = a)$.

A good method gives a matrix with large diagonal entries. Off-diagonal mass shows model mimicry: one model can produce data that look like another's.

What we actually need when interpreting real data is the reverse: given that model b won, how likely is it that b generated the data? This is the inversion matrix $P(\text{generated} = a \mid \text{selected} = b)$, obtained by normalizing each *column* of the confusion counts (assuming equal prior frequency of the generating models).

> Note: model recovery also reveals biases of criteria: BIC tends to favor simpler models at small N, AIC more complex ones. The "best" criterion is the one that recovers your candidate models best *at your design*, which is exactly what this study measures.

Reading the math

- Confusion matrix $P(\text{selected} = b \mid \text{generated} = a)$: rows are the true (generating) models, columns the selected ones. Each row adds up to 1. Read a row as: "when the truth is a, how often do we pick each model?"
- Inversion matrix $P(\text{generated} = a \mid \text{selected} = b)$: the same counts, but each *column* is normalized to 1. Read a column as: "when we picked b, how often was b really the truth?" This is the question that matters for a real dataset, where we only see the winner.
- Small example. 100 datasets from each of two models. From A: 90 select A, 10 select B. From B: 40 select A, 60 select B. Confusion row A = (0.90, 0.10). But among the 130 datasets that selected A, only 90 came from A: $P(\text{A true} \mid \text{A selected}) = 90/130 = 0.69$.

5. Identifiability

- Structurally non-identifiable: different parameter values give *exactly* the same predictions, for any data. Examples: label switching in the two-trace model (Week 3), or the SDT location and scale (Week 2). The fix is a constraint or a reparameterization.
- Practically non-identifiable: parameters are identifiable in principle but not with the available data. Recovery is poor and estimates trade off. The fix is a better design or a simpler model.

Parameter recovery is the practical test of identifiability.

For an engineer

- AIC and BIC are the model-order selection criteria you may know from AR/ARMA modeling; BIC is essentially Rissanen's MDL criterion. The LRT for nested models is again the GLRT.
- Parameter recovery is testing an estimator on synthetic signals with known ground truth, at the real SNR and record length, before trusting it on measured data. Engineers do this routinely; in psychology it is often skipped, which is why the course insists on it.
- The confusion matrix is the channel transition matrix of a discrete channel: input = generating model, output = selected model. The inversion matrix is the matrix of *a posteriori* probabilities $P(\text{input} \mid \text{output})$, i.e. what an MAP decoder needs, assuming equal input priors.
- Structural non-identifiability is like an unobservable state or a rank-deficient system matrix: no amount of data helps. Practical non-identifiability is an ill-conditioned problem: identifiable in principle, but the noise is too large for the available record.

For the instructor

- Frame recovery as "checking the equipment". No one would trust a thermometer that was never tested in water of known temperature. Recovery is that test for a model.
- Show one scatter plot of true vs. recovered values (lab Part C) before any theory: good recovery (points on the diagonal) next to bad recovery (a flat cloud). The picture explains itself.
- Teach confusion vs. inversion with the small 2×2 example above; students confuse them in the exam more than anything else this week. The medical-test analogy helps: the sensitivity of a test is not the probability that a positive patient is ill.
- Typical questions.

- *"AIC or BIC?"* Depends on the goal (prediction vs. finding the true model) and, practically, on which one recovers your models better at your design.
- *"How many simulated datasets?"* 50–100 per condition is enough to see the pattern; more for publication.
- *"Recovery failed, what now?"* Do not interpret that parameter. Change the design or simplify the model.

1.1.5 Worked example

The four forgetting models fitted by MLE to the Week 2 data (training NLLs from Week 6), with $N = 400$ item responses:

Model	k	NLL	AIC $= 2\,\text{NLL} + 2k$	BIC $= 2\,\text{NLL} + k \ln 400$
exp	2	29.812	63.62	71.61
pow	2	19.830	43.66	51.64
expasym	3	18.919	43.84	55.81
two	4	18.919	45.84	61.80

Step 1: AIC for the power model. $2 \times 19.830 + 2 \times 2 = 43.66$.

Step 2: BIC for the power model. $\ln 400 = 5.991$, so $2 \times 19.830 + 2 \times 5.991 = 51.64$.

Step 3: Akaike weights. Differences from the best AIC: $\Delta = (19.96, 0, 0.18, 2.18)$. Then $e^{-\Delta/2} = (0.00005, 1, 0.914, 0.336)$, which sum to 2.250, so

$$w^{\text{AIC}} = (0.000,\ 0.444,\ 0.406,\ 0.149).$$

BIC weights computed the same way: $(0.000,\ 0.884,\ 0.110,\ 0.005)$.

Step 4: likelihood-ratio test, exp nested in expasym: $\Lambda = 2(29.812 - 18.919) = 21.79$ on 1 df, $p \approx 3 \times 10^{-6}$. Adding an asymptote improves the exponential model significantly.

Interpretation. Both criteria reject the pure exponential decisively. AIC cannot separate the power and asymptote models (weights 0.44 vs. 0.41): for *prediction* they are about equally good, consistent with the Week 6 CV results. BIC's larger penalty favors the two-parameter power model (0.88). The two-trace model's extra parameter buys nothing: identical NLL to expasym, so it pays the penalty for nothing. Whether we should *trust* these preferences at this design is a question for model recovery (lab Part D).

1.1.6 Lab (2 hours)

Goals

1. Run a likelihood-ratio test and compute AIC, AICc, BIC and weights for the forgetting models.
2. Run a parameter-recovery study for the power and exponential-with-asymptote models, and check for trade-offs.
3. Run a model-recovery study for three models with AIC and BIC; build confusion and inversion matrices.
4. Interpret: what can and cannot be concluded from the Week 2 data at this design?

Timing: Part A 15 min · Part B 15 min · Part C 35 min · Part D 40 min · Part E 15 min.

Setup

Code · کد: `code/week07.py`, cell 1 · سلول 1

Part A: likelihood-ratio test (15 min)

Code · کد: `code/week07.py`, cell 2 · سلول 2

Expected output:

```
NLL exp = 29.812, NLL expasym = 18.919
LR statistic = 21.79, df = 1, naive p = 3.05e-06, boundary-corrected p =
↪  1.52e-06
```

Question: why can't you use the same test to compare the power and exponential models?

Part B: information criteria and weights (15 min)

Code · کد: `code/week07.py`, cell 3 · سلول 3

Expected output:

```
         k     NLL     AIC    AICc     BIC  w_AIC  w_AICc  w_BIC
model
exp      2  29.812  63.625  63.655  71.608  0.000   0.000  0.000
pow      2  19.830  43.661  43.691  51.644  0.444   0.449  0.884
expasym  3  18.919  43.837  43.898  55.812  0.406   0.405  0.110
two      4  18.919  45.837  45.939  61.803  0.150   0.146  0.006
```

Sensitivity check: recompute BIC with $N = 8$ (the number of cells) instead of 400. Does the ranking change? Which choice of N is defensible here, and why?

Part C: parameter recovery (35 min)

Code · کد: `code/week07.py`, cell 4 · سلول 4

Expected output (about 1 minute):

```
pow:
   r(true, rec)   bias   RMSE
A         0.888  0.005  0.055
d         0.965  0.003  0.070

expasym:
     r(true, rec)   bias   RMSE
A           0.802  0.008  0.077
lam         0.799  0.034  0.230
s           0.918 -0.008  0.060

expasym: corr(true lam, true s) = -0.08;  corr(recovered lam, recovered
↪  s) = 0.03
```

Interpret: 1. Which parameters are well recovered at this design? Which would you hesitate to interpret? 2. The expasym decay rate λ has by far the largest RMSE (0.23 on a range of 0.05–1). Look at its panel: for which true values does recovery break down, and why? (Hint: if decay is very fast, most of the decline happens before the second test interval, so what do the data look like?) Here the recovered λ and s are *not* correlated ($r = 0.03$), so this is imprecision rather than a trade-off. Contrast this with the clear A–d trade-off of the power model found in Week 5. 3. What design change would improve recovery of λ and s? Test one idea in Exercise 3.

Part D: model recovery (40 min)

Code · کد: `code/week07.py`, cell 5 · سلول 5

Expected output (about 1–2 minutes):

```
AIC confusion matrix (rows = generating model, columns = selected model):
          exp   pow  expasym
exp      0.91  0.00     0.09
pow      0.03  0.82     0.15
expasym  0.06  0.25     0.69
AIC inversion matrix (rows = generating model, columns = selected model):
          exp   pow  expasym
exp      0.91  0.00     0.10
pow      0.03  0.77     0.16
expasym  0.06  0.23     0.74

BIC confusion matrix (rows = generating model, columns = selected model):
          exp   pow  expasym
exp      0.99  0.01     0.00
pow      0.05  0.94     0.01
expasym  0.09  0.58     0.33
BIC inversion matrix (rows = generating model, columns = selected model):
          exp   pow  expasym
exp      0.88  0.01     0.00
pow      0.04  0.61     0.03
expasym  0.08  0.38     0.97
```

Interpret: 1. Which model is most often confused with which? (With AIC, 25% of expasym datasets are labeled power; with BIC, 58%.) Explain the mimicry in terms of the curves' shapes. 2. Compare AIC and BIC. Which recovers the simpler models better, and which the more complex one? 3. In Part B, BIC selected the power model for the real data. Using the BIC inversion matrix, how confident should we be that the data were generated by a power model? (Only 61% of BIC's "power" verdicts came from power-generated data; 38% came from expasym.)

Part E: conclusions about the Week 2 data (15 min)

Write a paragraph (6–10 sentences) that a reviewer would accept, summarizing what the Week 2 data say about the form of forgetting. Use Parts A–D and your Week 6 CV results, and state what *cannot* be concluded at this design.

Exercises (increasing difficulty)

Exercise 1 $\star$. Compute AIC and BIC weights for the worked example by hand (calculator) and check them against Part B.

Exercise 2 $\star$. Repeat Part B with AICc. For which N would AICc and AIC differ noticeably for the two-trace model?

Exercise 3 $\star\star$ (design for recovery). Rerun the expasym recovery of Part C with intervals extended to $\{0.25, 1, 2, 4, 7, 14, 30, 60, 90\}$ days, and separately with $n = 200$ items. Which change helps λ and s recovery more?

Exercise 4 ⋆⋆ (CV in model recovery). Replace AIC/BIC in Part D by 5-fold CV (from Week 6) for 30 simulations per model. Is CV better or worse than AIC at recovering the models? Is it worth the computation time?

Exercise 5 ⋆⋆⋆ (Advanced: SDT recovery). For the payoff experiment of Week 4 (three conditions, 80 + 80 trials per condition), run a model recovery for "shared d'" vs. "separate d'" using the likelihood-ratio test at $\alpha = .05$ and BIC. How often does each method wrongly claim that payoffs change sensitivity? How many trials per condition would you need to detect a true sensitivity difference of 0.3 in 80% of simulations?

Lab check-off

Show: (1) your IC table with weights (Part B), (2) the recovery scatter plots with one sentence on which parameter is least well recovered and why, (3) the AIC and BIC confusion matrices, and (4) your Part E paragraph.

1.1.7 Connection to AI & LLMs

Comparing learning curves of an AI model. LLMs can learn a new task from a few examples in the prompt ("in-context learning"). How the error falls with the number of examples k is a learning curve, and the question "exponential or power?" is the same question as in this week's lab (and Week 2's forgetting curves). Information criteria, model recovery and identifiability apply unchanged. The design matters as much as before: with few values of k and few test items, the two curves may be impossible to tell apart.

Code · كد: `code/week07.py`, cell 6 · سلول 6

Expected output:

```
{'exponential': 41.6, 'power': 40.4} Akaike weights: [0.362 0.638]
```

The data were generated by a power law, but the evidence is weak (weights 0.64 vs. 0.36).

Try it (no API needed). - ⋆ Run a model recovery (Week 7, Part D) at this design: 50 datasets from each curve. Build the confusion matrix. - ⋆⋆ Change the design: more values of k (up to 128) or 200 test items per k. Which change improves model recovery more? - ⋆⋆ Discussion: a paper concludes from one fitted curve that "in-context learning follows a power law, like human practice". What would you need to see before accepting the comparison with humans?

1.1.8 Common mistakes & misconceptions

1. Interpreting an AIC value on its own. Only differences between models fitted to the same data mean anything.
2. Using LRTs for non-nested models, or reading a non-significant LRT as evidence *for* the simpler model.
3. Thinking BIC is "Bayesian AIC" with the same goal. BIC aims at identifying the data-generating model; AIC aims at prediction. They can disagree for good reasons.
4. Unclear N. Always report what counted as an observation; BIC and AICc depend on it.
5. Skipping recovery. A parameter that cannot be recovered at your design must not be interpreted, no matter how good the fit.
6. Running recovery at a different design or with a different fitting procedure than the real analysis. Recovery results are design-specific.
7. Reading the confusion matrix as if it were the inversion matrix. "Data from model A are recovered as A 90% of the time" is not the same as "when A wins, it is the true model 90% of the time".
8. Choosing parameter ranges for recovery that are wider or narrower than plausible values. Too wide inflates correlations; too narrow hides problems.

1.1.9 Check your understanding

1. Write the formulas for AIC and BIC. For which N does BIC penalize each parameter more than AIC? Answer AIC $= -2\hat{\ell} + 2k$; BIC $= -2\hat{\ell} + k \ln N$. BIC's penalty exceeds AIC's when $\ln N > 2$, i.e., $N > e^2 \approx 7.4$, so for $N \geq 8$.

2. Model A has AIC = 210.4, model B has AIC = 212.9. Compute the Akaike weights and interpret them. Answer $\Delta = (0, 2.5)$, so $e^{-\Delta/2} = (1, 0.287)$ and $w = (0.777, 0.223)$. A is favored about 3.5:1 as the better predictor, but B retains non-negligible support.

3. Why can the likelihood-ratio test compare exponential and exponential-with-asymptote but not exponential and power? Answer The exponential is a special case of the asymptote model (fix $s = 0$), so the models are nested and $2\Delta\ell$ has a known χ^2 distribution under the null. Exponential and power are not nested: neither is a special case of the other.

4. Describe the steps of a parameter-recovery study. Answer Choose plausible true parameter values; simulate data at the real design; fit with the real fitting procedure; compare recovered with true values (scatter plots, correlation, bias, RMSE); check correlations among the recovered parameters for trade-offs.

5. In a recovery study, the recovered λ and s correlate at $r = -0.8$, although the true values were drawn independently. What does that mean? Answer A trade-off: at this design, different combinations of decay rate and asymptote produce nearly the same predictions, so errors in one are compensated by the other. Individual values of λ and s should not be interpreted separately. A better design or a reparameterization is needed.

6. Explain the difference between a confusion matrix and an inversion matrix in model recovery. Answer Confusion: $P(\text{selected} \mid \text{generated})$, with rows summing to 1, which describes the method's behavior for each true model. Inversion: $P(\text{generated} \mid \text{selected})$, with columns summing to 1 (assuming equal priors for the generating models), which tells us how much to trust a given winner when analyzing real data.

7. A model recovery shows that AIC selects the three-parameter model 40% of the time when the two-parameter model generated the data. What would you do? Answer Treat AIC wins for the complex model with caution at this design: check BIC or CV recovery, which may be more conservative, improve the design (more trials, more informative conditions), and report the recovery results alongside the real-data comparison.

1.1.10 Assignment

A3 continues (due at the start of the Week 9 lecture)

Tasks 4 (information criteria) and 5 (parameter and model recovery) of A3 use this week's material. The full brief is in Week 6. Reuse today's `parameter_recovery` code and the model-recovery loop: they are designed to be adapted directly.

Final project: teams and proposal brief released

- Teams: 1–2 students. Register your team with the TA by the end of this week.
- Task: choose a cognitive model and apply the full pipeline to a dataset (simulated, open, or your own): fit, evaluate (at least one model comparison with an alternative model), recover (at least one parameter-recovery analysis), interpret. Pairs also include a Bayesian/MCMC or hierarchical analysis and an individual contribution statement.
- Proposal (due at the start of the Week 9 lecture, max. 2 pages): research question, model(s) with equations, data source and design, planned comparison and recovery analyses, and a timeline. The full project brief and rubric are distributed separately (`project_brief.md`).
- Not sure what to model? The topics of Weeks 14–15 (exemplar categorization, drift-diffusion, reinforcement learning) all make good projects, as do extensions of the forgetting and SDT models from this course.

1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapter 10. Model comparison: likelihood-ratio tests, AIC, BIC and their weights; their assumptions and limitations. *(Main reading.)*
- Lee & Wagenmakers (2014), chapter 7 (introduction). How the Bayesian approach to model comparison relates to the methods of this week, as preparation for Week 10.

- *Optional, outside the course list:* Wilson, R. C., & Collins, A. G. E. (2019). Ten simple rules for the computational modeling of behavioral data. *eLife, 8*, e49547. A practical guide that treats parameter and model recovery as standard practice.

Preview for Week 8: the midterm. The lecture is an integrated review in which we run the whole pipeline (simulate, fit, evaluate, recover, interpret) on a new dataset; the lab slot is the midterm exam.